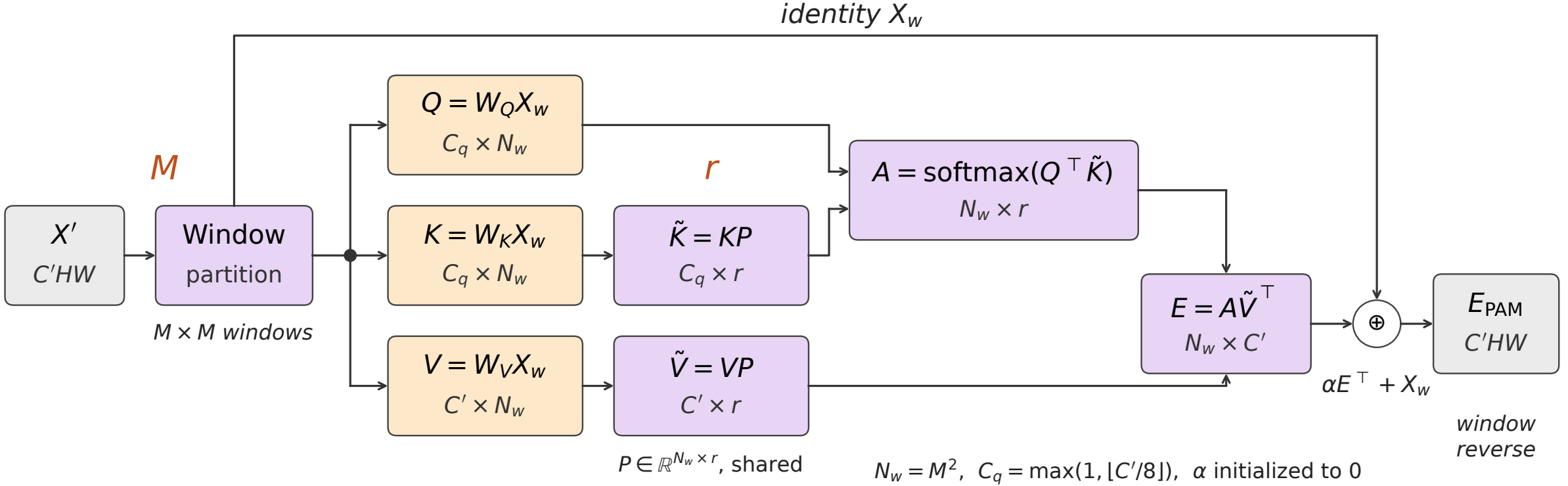
identity $X_w$
$M$
$r$
$X'$
$C'HW$
Window
partition
$M \times M$ windows
$Q = W_Q X_w$
$C_q \times N_w$
$K = W_K X_w$
$C_q \times N_w$
$V = W_V X_w$
$C' \times N_w$
$\tilde{K} = KP$
$C_q \times r$
$\tilde{V} = VP$
$C' \times r$
$P \in \mathbb{R}^{N_w \times r}$, shared
$A = \mathrm{softmax}(Q^\top \tilde{K})$
$N_w \times r$
$E = A\tilde{V}^\top$
$N_w \times C'$
$\alpha E^\top + X_w$
$E_{\mathrm{PAM}}$
$C'HW$
window reverse
$N_w = M^2$, $C_q = \max(1, \lfloor C'/8 \rfloor)$, $\alpha$ initialized to 0